AI-Based Study Planner Using a Rational Agent

1. Problem Formulation

Students often find it difficult to manage multiple subjects, deadlines and limited study time. This project develops an AI-Based Study Planner that automatically creates a weekly study schedule.

The agent takes the following inputs:

Subject name

Deadline

Difficulty level

Priority or importance

Available daily study hours

The goal is to create a rational study plan by giving higher importance to urgent and high-priority subjects.

2. Approach

The AI Study Planner uses a priority-based approach. Each subject is evaluated using factors such as deadline urgency, difficulty and priority.

A higher utility score is given to subjects that are more urgent and important. The planner then schedules these subjects first based on the available study hours.

3. Rational Agent Approach

This project acts as a Rational Agent.

Perceive: The agent receives subject details and available

study hours.

Decide: The agent calculates the priority or utility score for each subject.

Act: The agent generates a weekly study timetable by scheduling high-priority subjects first.

This approach is rational because the agent selects actions that help achieve the objective of completing important subjects before their deadlines.

4. Complexity Analysis

Let n be the number of subjects.

The system calculates the priority for each subject and arranges the subjects based on their priority.

The approximate time complexity is:

O(n log n)

The space complexity is:

O(n)

5. Results

The AI-Based Study Planner successfully accepts subject details and available study hours. It calculates a utility score and generates a readable weekly study schedule.

The program was tested using subjects with different deadlines and difficulty levels. Subjects with higher urgency and priority were scheduled first.

```
Enter the number of subjects: 2
Enter available study hours per day: 3

Enter details for Subject 1
Subject name: AI
Deadline (YYYY-MM-DD): 2026-10-25
Difficulty (1 = Easy, 5 = Very Hard): 1
Priority/Weight (1 = Low, 5 = Very High): 5

Enter details for Subject 2
Subject name: JAVA
Deadline (YYYY-MM-DD): 2026-10-26
Difficulty (1 = Easy, 5 = Very Hard): 1
Priority/Weight (1 = Low, 5 = Very High): 5

============================================================
          AI-BASED WEEKLY STUDY PLANNER
============================================================

Objective: Maximize coverage of urgent and high-priority topics before deadli
Available Study Hours Per Day: 3.0

------------------------------------------------------------
Sunday, 06-09-2026
------------------------------------------------------------
AI | 1 hour(s) | Utility Score: 1.02 | Deadline: 2026-10-25
JAVA | 1 hour(s) | Utility Score: 1.0 | Deadline: 2026-10-26

------------------------------------------------------------
Monday, 07-09-2026
------------------------------------------------------------
AI | 1 hour(s) | Utility Score: 1.02 | Deadline: 2026-10-25
JAVA | 1 hour(s) | Utility Score: 1.0 | Deadline: 2026-10-26

------------------------------------------------------------
Tuesday, 08-09-2026
```

```
------------------------------------------------------------
📚 AI | 1 hour(s) | Utility Score: 1.02 | Deadline: 2026-10-25
📚 JAVA | 1 hour(s) | Utility Score: 1.0 | Deadline: 2026-10-26

------------------------------------------------------------
Wednesday, 09-09-2026
------------------------------------------------------------
📚 AI | 1 hour(s) | Utility Score: 1.02 | Deadline: 2026-10-25
📚 JAVA | 1 hour(s) | Utility Score: 1.0 | Deadline: 2026-10-26

------------------------------------------------------------
Thursday, 10-09-2026
------------------------------------------------------------
📚 AI | 1 hour(s) | Utility Score: 1.02 | Deadline: 2026-10-25
📚 JAVA | 1 hour(s) | Utility Score: 1.0 | Deadline: 2026-10-26

------------------------------------------------------------
Friday, 11-09-2026
------------------------------------------------------------
📚 AI | 1 hour(s) | Utility Score: 1.02 | Deadline: 2026-10-25
📚 JAVA | 1 hour(s) | Utility Score: 1.0 | Deadline: 2026-10-26

------------------------------------------------------------
Saturday, 12-09-2026
------------------------------------------------------------
📚 AI | 1 hour(s) | Utility Score: 1.02 | Deadline: 2026-10-25
📚 JAVA | 1 hour(s) | Utility Score: 1.0 | Deadline: 2026-10-26


============================================================
WHY IS THIS AGENT RATIONAL?
============================================================

The agent follows a utility-based rational decision-making process.

For every subject, it considers:
```

```
============================================================
WHY IS THIS AGENT RATIONAL?
============================================================

The agent follows a utility-based rational decision-making process.

For every subject, it considers:

1. Priority Weight
   - Important subjects receive higher priority.

2. Difficulty
   - Difficult subjects require more preparation.

3. Deadline Urgency
   - Subjects with closer deadlines become more urgent.

The utility function combines these factors:

Utility = (Priority x Difficulty x 10) / Remaining Days

The agent selects subjects with the highest utility scores first.
Therefore, the generated timetable attempts to maximize the
coverage of important and urgent subjects within the available
daily study hours.
```

6. Reflection

The main challenge was designing a priority rule that

considers different factors such as deadlines and difficulty.

In the future, the project can be improved by adding:

A graphical user interface

Automatic reminders

More detailed subject analysis

Calendar integration

7. Honesty Note

AI assistance was used for guidance and learning during the development of this project. The final project was reviewed and understood before submission.

8. Conclusion

The AI-Based Study Planner demonstrates how a Rational Agent can make intelligent decisions based on available information. The project helps students organise their study time by prioritising urgent and important subjects.